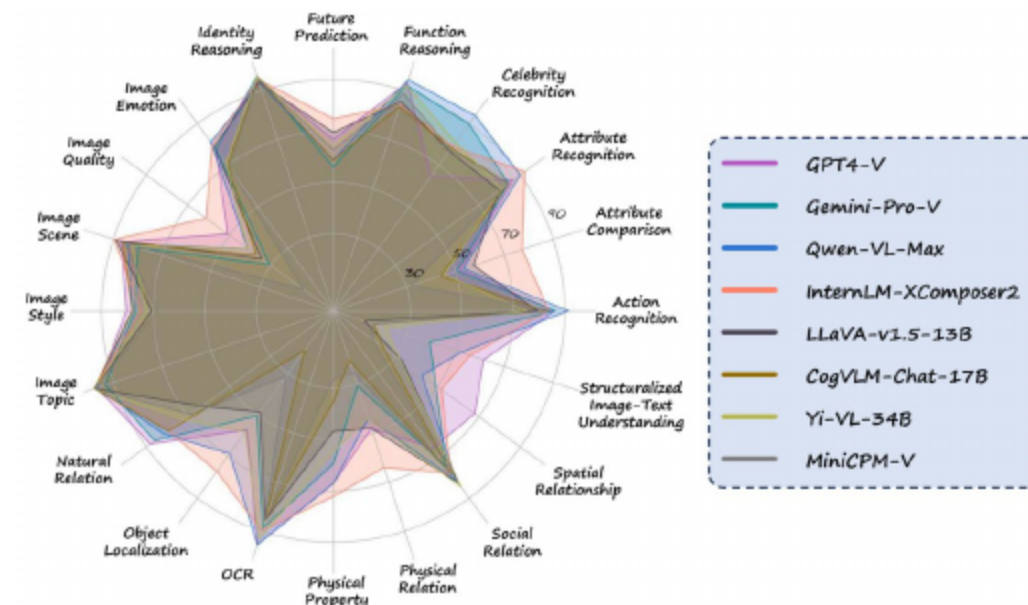


MMBench: Is Your Multi-modal Model an All-around Player?

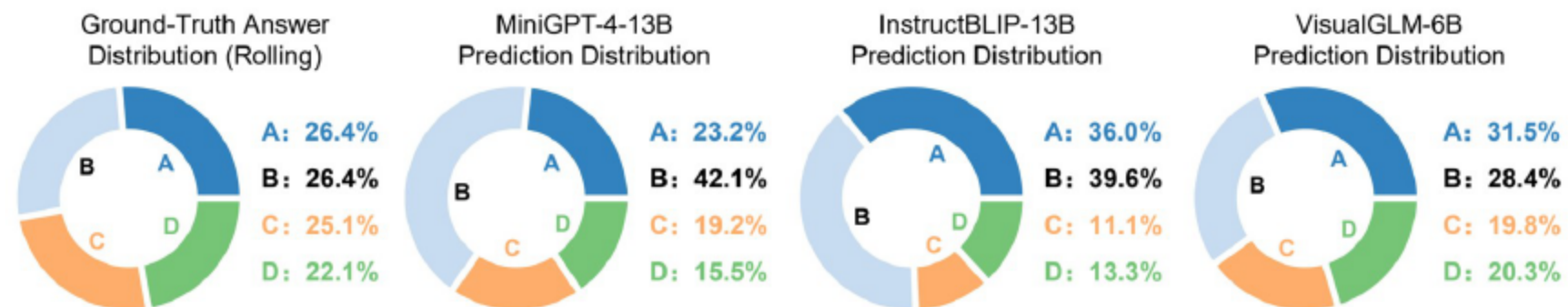
Yuan Liu; Haodong Duan; Yuanhan Zhang; Bo Li; Songyang Zhang; Wangbo Zhao; Yike Yuan; Jiaqi Wang; Conghui He; Ziwei Liu; Kai Chen; Dahua Lin

01 Need for Rigorous LVLM Evaluation



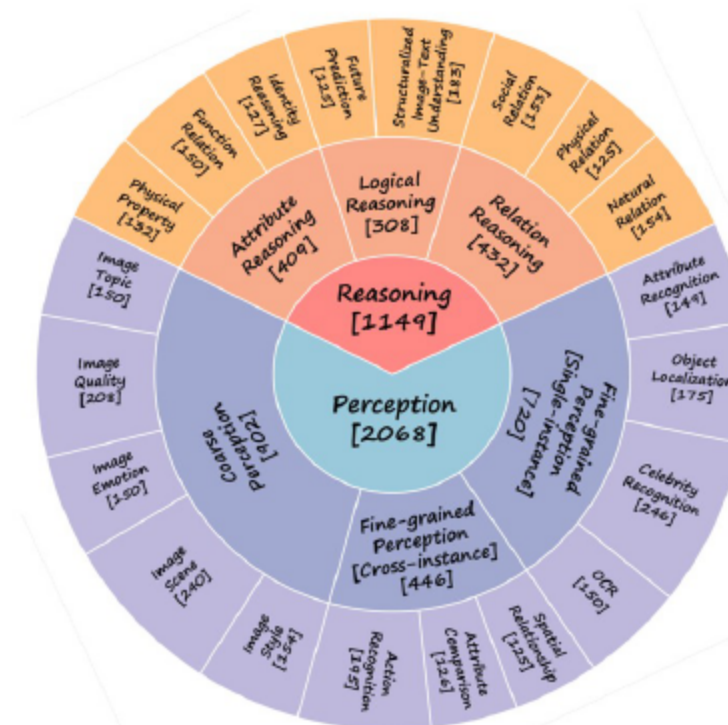
- Recent LLMs (e.g., GPT-4) show strong reasoning; LVLMs (GPT-4v, Gemini-Pro-V, LLaVA) inherit **promising multimodal abilities**
- Early studies favored qualitative demos, limiting fair comparison across models
- Gap:** lack of rigorous, quantitative, fine-grained evaluation for LVLMs hampers progress

02 Limitations of Existing Benchmarks



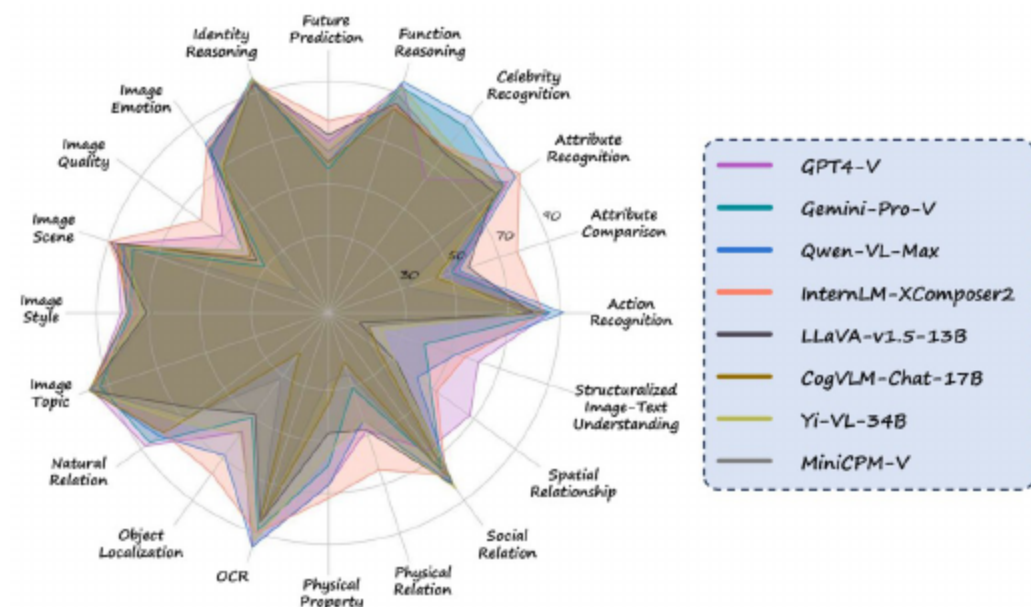
- Objective datasets: exact-match scoring creates **false negatives** (e.g., bicycle vs bike)
- Narrow task focus restricts **fine-grained capability** analysis
- Subjective evaluations suffer **bias**, low reproducibility, and high cost; limited scalability

03 Evaluation



- Dataset:** 3,217 MCQs covering 20 abilities; balanced counts per ability for stable comparisons
- LLM-assisted choice extraction reduces **matching errors**; GPT-4 aligns with humans strongly
- CircularEval increases **robustness** and exposes performance gaps beyond single-pass

04 Benchmarks and Model Landscape: Scope and Gaps



- Existing datasets cover captions, VQA, OCR, science QA, or video, but use heterogeneous metrics
- Problem:** task-specific scope prevents **holistic**, ability-level assessment
- Diverse open and proprietary VLMs require a unified, fine-grained, **reproducible** benchmark

[Fu et al. 2023; Singh et al. 2019]

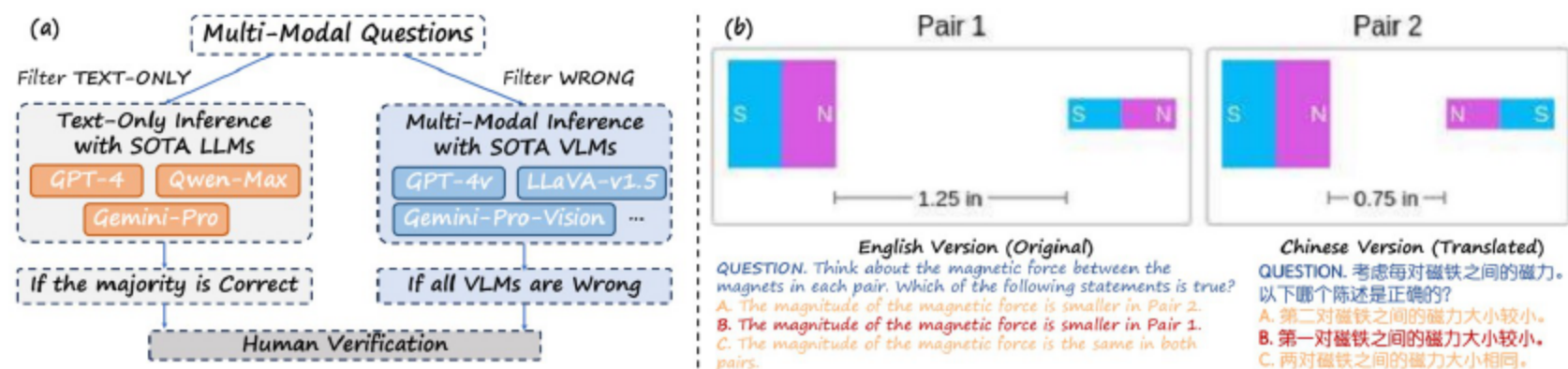
05 Goals and Ability Taxonomy



- Hierarchical taxonomy: Perception and Reasoning → six level-2 groups → twenty level-3 abilities
- Images and problems from varied sources; **bilingual** English and Chinese for apple-to-apple comparison
- Clear ability definitions enable **diagnostic** per-ability reporting

06 Collection Pipeline and Quality Control

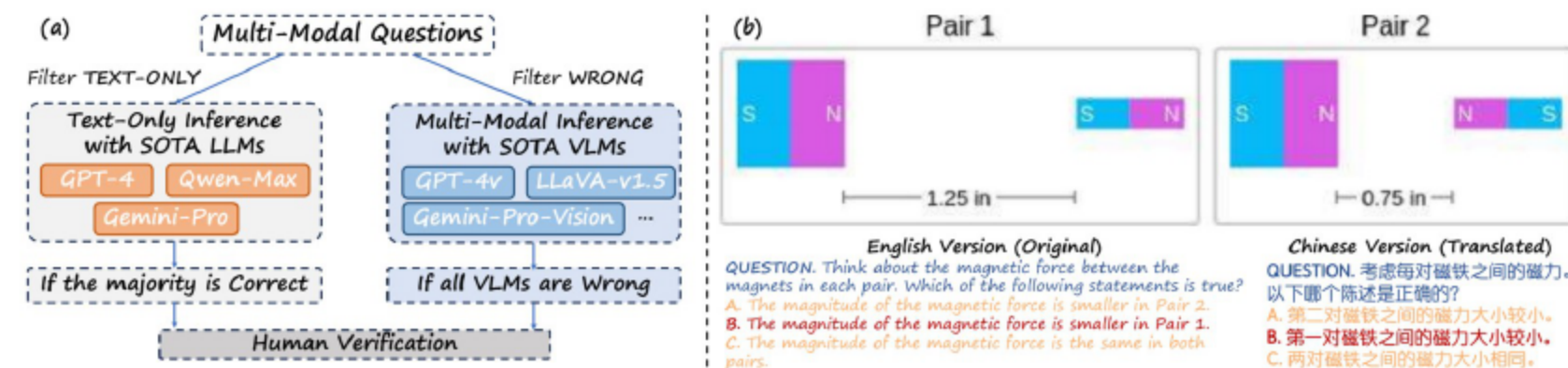
6 Yuan Liu et al.



- Volunteer-curated MCQs per ability; majority-vote text-only filter using multiple LLMs
- Automatic wrong-sample screening with multiple VLMs plus manual verification
- Quality control removes text-only leakage and **erroneous** items, improving dataset fidelity

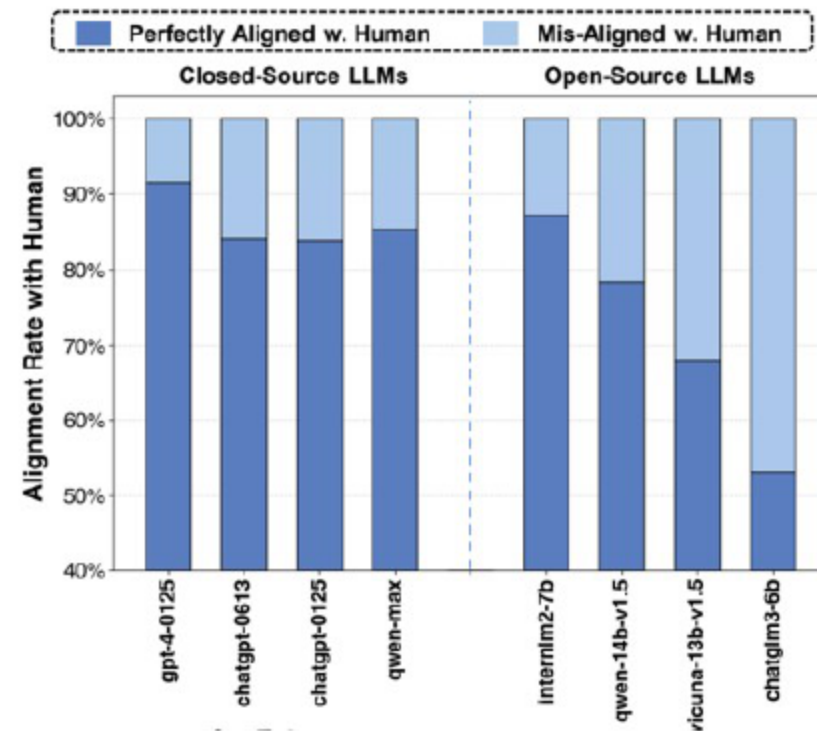
07 Dataset Statistics, Splits, and Bilingual Version

6 Yuan Liu et al.



- 3,217 samples across 20 abilities; maintain roughly even per-ability counts for stability
- Standard 4:6 dev/test split; dev answers public, test via evaluation server for **fairness**
- MMBench-CN translated by GPT-4 and human-verified; supports **bilingual** evaluation

- Heuristic label parsing first; fallback to an LLM (gpt-4-0125) to align free-form outputs to choices
- Extractor returns A, B, C, or D; outputs Z when no option matches to avoid **spurious** credit
- **Outcome:** improved matching reduces **false negatives** and standardizes scoring across VLMs



The original VL problem:

Q: How many apples are there in the image?
A. 4; B. 3; C. 2; D. 1
GT: A

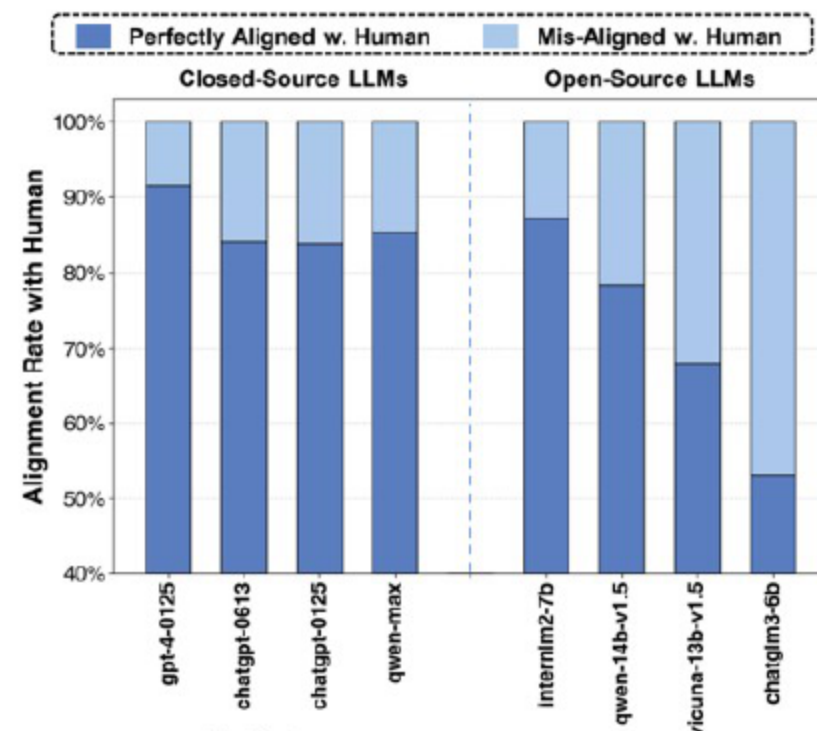
Circular Evaluation

4 Passes in Circular Evaluation (choices with circular shift):

1. Q: How many apples are there in the image? Choices: A. 4; B. 3; C. 2; D. 1. VLM prediction: A. GT: A ✓
 2. Q: How many apples are there in the image? Choices: A. 3; B. 2; C. 1; D. 4. VLM prediction: D. GT: D ✓
 3. Q: How many apples are there in the image? Choices: A. 2; B. 1; C. 4; D. 3. VLM prediction: B. GT: C ✗
 4. Q: How many apples are there in the image? Choices: A. 1; B. 4; C. 3; D. 2. VLM prediction: B. GT: B ✓
- VLM failed at pass 3. Thus wrong.

- Run multiple passes with circular-shifted choices; a sample counts only if all passes are correct
- Mitigates guessing and choice-position bias; early stop keeps cost practical
- Produces **more discriminative** results and reveals hidden robustness gaps

- Zero-shot with a unified prompt; open-ended generation; GPT-4 used as the choice extractor
- Model coverage: text-only GPT-4, many open-source VLMs, and proprietary VLMs
- Consistent settings ensure **fair**, reproducible comparisons across architectures and scales



VLM	Circular	Acc Change	VLM	Circular	Acc Change	VLM	Circular	Acc Change
MiniGPT4-7B	32.7%	-24.1%	MiniGPT4-13B	37.5%	-23.2%	Yi-VL-6B	65.6%	-9.8%
InstructBLIP-7B	37.4%	-24.0%	InstructBLIP-13B	40.9%	-23.0%	Yi-VL-34B	68.2%	-9.5%
LLaVA-v1.5-7B	62.5%	-11.2%	LLaVA-v1.5-13B	67.2%	-8.6%	MiniCPM-V	64.8%	-10.6%
IDEFICS-9B-Instruct	37.2%	-22.6%	LLaVA-InternLM2-20B	72.8%	-7.0%	Qwen-VL-Plus	62.9%	-16.6%
VisualGLM-6B	36.1%	-27.0%	CogVLM-Chat-17B	62.4%	-15.6%	Qwen-VL-Max	76.4%	-8.7%
Qwen-VL-Chat	59.5%	-17.4%	mPLUG-Owl2	63.5%	-8.7%	Gemini-Pro-V	70.9%	-11.7%
OpenFlamingo v2	2.6%	-34.1%	InternLM-XComposer2	79.1%	-4.7%	GPT-4v	74.3%	-10.8%

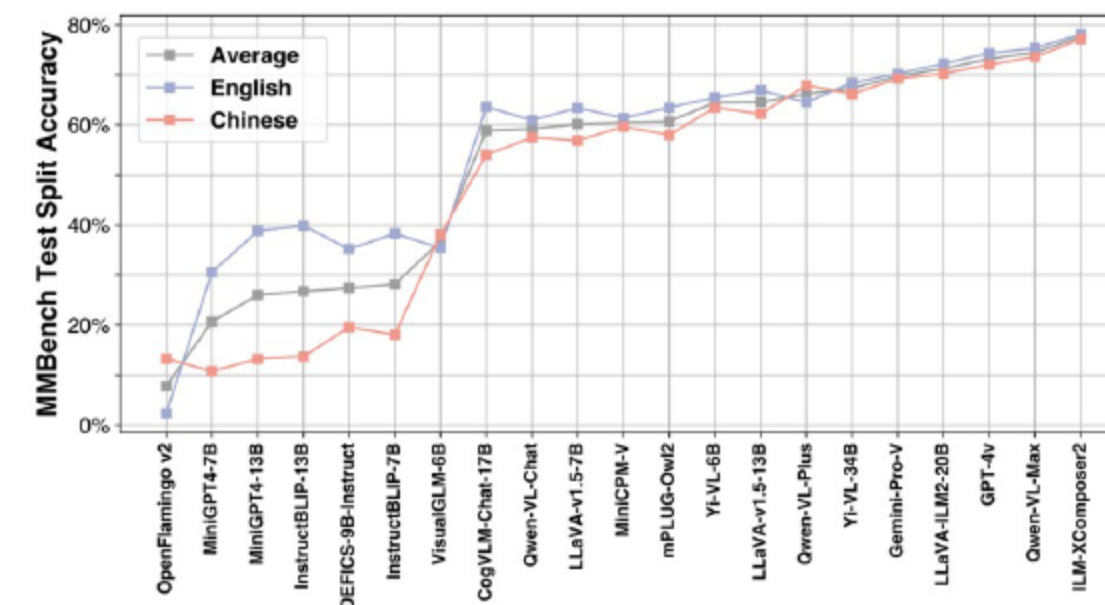
- Switching to CircularEval leads to notable accuracy drops, exposing **stability** deficits
- Performance gaps between model sizes and families become **clearer** under circular testing
- Single-pass scores can be **over-optimistic**; circular consistency is more reliable

12 Key Findings and Per-Ability Highlights

Model	Overall	CP	FP-S	FP-C	AR	LR	RR
Large Language Models							
GPT-4-Turbo (0125) [26]	2.9%	0.6%	1.2%	4.1%	3.7%	4.9%	7.4%
OpenSource VLMs							
OpenFlamingo v2 [3]	2.3%	1.1%	3.5%	1.5%	5.3%	0.0%	2.7%
MiniGPT4-7B [41]	30.5%	37.0%	31.8%	17.2%	49.8%	9.2%	25.6%
IDEFICS-9B-Instruct [19]	35.2%	48.3%	31.3%	29.6%	47.8%	11.4%	25.2%
VisualGLM-6B [11]	35.4%	40.2%	38.5%	26.2%	47.8%	19.6%	29.5%
InstructBLIP-7B [9]	38.3%	46.7%	39.0%	31.8%	55.5%	8.7%	31.0%
MiniGPT4-13B [41]	38.8%	44.6%	42.9%	23.2%	64.9%	8.2%	32.9%
InstructBLIP-13B [9]	39.8%	47.2%	42.9%	21.0%	60.4%	12.5%	38.8%
Qwen-VL-Chat* [4]	60.9%	68.5%	67.7%	50.2%	78.0%	37.0%	45.7%
MiniCPM-V [27]	61.4%	65.0%	69.4%	51.3%	70.6%	35.3%	59.7%
LLaVA-v1.5-7B [21]	63.4%	70.0%	68.0%	57.7%	77.6%	33.2%	56.2%
mPLUG-Owl2 [37]	63.5%	68.1%	69.1%	55.8%	78.4%	37.0%	57.0%
CogVLM-Chat-17B [34]	63.6%	72.8%	66.6%	55.4%	71.4%	33.7%	62.0%
Yi-VL-6B* [1]	65.5%	72.8%	72.9%	56.2%	75.5%	41.3%	55.4%
LLaVA-v1.5-13B [21]	66.9%	73.1%	72.4%	60.3%	75.5%	35.9%	65.5%
Yi-VL-34B* [1]	68.4%	72.0%	78.0%	54.7%	81.2%	38.6%	68.2%
LLaVA-InternLM2-20B [8]	72.3%	78.3%	76.6%	68.2%	78.4%	46.2%	69.4%
InternLM-XComposer2* [10]	78.1%	80.4%	83.5%	73.0%	83.7%	63.6%	74.4%
Proprietary VLMs							
Qwen-VL-Plus [4]	64.6%	66.5%	79.1%	50.2%	73.9%	42.9%	57.8%
Gemini-Pro-V [31]	70.2%	70.0%	78.9%	65.9%	82.9%	46.2%	65.9%
GPT-4v [26]	74.3%	77.6%	73.8%	71.5%	85.3%	63.6%	68.6%
Qwen-VL-Max [4]	75.4%	74.8%	87.2%	67.0%	85.3%	54.9%	70.5%

- Among open-source, InternLM-XComposer2 ranks highest; LLaVA and Yi-VL variants are strong
- LLM choice and scaling improve **reasoning** and overall robustness
- Proprietary vs open-source: proprietary excel on structured inputs and external-knowledge tasks

13 Bilingual Evaluation and Content Moderation

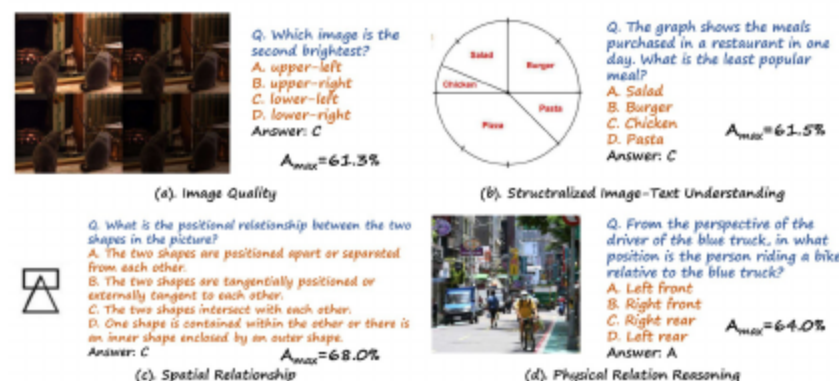


- Most models slightly worse on Chinese; top models show **small** English-Chinese gaps
- Content moderation causes **minor** accuracy penalties (up to about 2.4 percent)
- Rejections cluster on celebrity recognition for some APIs; limited impact on overall ranking

14 Hard Cases and Open Challenges



Fig. 9: Proprietary VLMs vs. Open-Source ones at a fine-grained level.



- Persistent weaknesses: low-level visual features (brightness, contrast, artifacts) are **error-prone**
- Structured inputs (tables, diagrams, layouts) remain challenging across models
- Inter-object spatial relations in 2D and 3D drive **systematic** failures under robust evaluation

THANKS!